

Hema Vardhan Moganti

Melbourne, VIC, Australia
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Cloud & AI Solutions Developer | Azure & Multi-Cloud | Cybersecurity

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PROFESSIONAL SUMMARY

Final-year Bachelor of Networking (Cyber Security) student with hands-on cloud and AI development experience spanning Microsoft Azure, AWS, and Google Cloud. Demonstrated ability to design, deploy, and manage cloud-native solutions — including an AI-powered multi-cloud text recognition and image enhancement system — and secure enterprise networks. Proficient in Python, Bash, and PowerShell scripting with practical exposure to CI/CD pipelines, Git workflows, Infrastructure as Code, and DevOps practices. Passionate about Azure PaaS services, proactive monitoring, automation, and delivering high-quality outcomes for customers in managed-services environments.

TECHNICAL SKILLS

- Cloud Platforms:** Microsoft Azure, AWS, Google Cloud Platform
- Azure Services:** Azure VM, Azure Functions, App Service, Azure Monitor, Log Analytics, Active Directory
- AWS Services:** Lambda, S3, Rekognition, EventBridge, CloudWatch
- GCP Services:** Vision API, Cloud Functions, Cloud Storage, Cloud Run
- Programming:** Python, Bash, PowerShell, HTML, CSS, SQL
- DevOps & IaC:** Git, GitHub Actions, CI/CD pipelines, Bicep/ARM (foundational), Infrastructure as Code
- AI / ML:** Computer Vision (OCR/text detection), OpenCV, Pillow, cloud-native AI APIs
- Networking & Security:** TCP/IP, DNS, DHCP, VPN, VLANs, Active Directory, Group Policy, Wireshark
- Tools:** Cisco Packet Tracer, Cisco Modelling Labs, Autopsy, FTK Imager, VS Code, PyCharm, SQLite, MS Access
- Operating Systems:** Windows Server 2022, Windows 10/11, Linux (Ubuntu)

EDUCATION

Bachelor of Networking (Major in Cyber Security)
Melbourne Institute of Technology · Melbourne, VIC 3000

Sep 2023 – Jun 2026
Expected Graduation

PROJECTS

AI-Powered Text Recognition & Image Enhancement System

AWS + Google Cloud

Designed and deployed a multi-cloud AI pipeline that automatically detects whether an uploaded image contains text, extracts that text via OCR, and — when the source image is blurred — applies image

enhancement algorithms before re-running extraction. Outputs are saved as separate enhanced image files and structured .txt files, enabling both human review and downstream automation.

- Leveraged **AWS Rekognition** and **Google Cloud Vision API** for dual-cloud text detection and OCR, improving accuracy through result reconciliation.
- Built serverless ingestion using **AWS Lambda + S3** triggers and **Google Cloud Functions + Cloud Storage** events for event-driven, auto-scaling processing.
- Implemented image pre-processing pipeline with **OpenCV** and **Pillow** — blur detection (Laplacian variance), sharpening filters, and contrast enhancement — before OCR to increase text extraction accuracy on low-quality images.
- Automated end-to-end workflow: image upload → AI text detection → conditional enhancement → OCR extraction → dual-output (enhanced image + .txt file) stored back in cloud storage.
- Integrated **CloudWatch** and **GCP Cloud Logging** for monitoring, alerting, and audit trails across both clouds.

Tools: AWS Lambda, S3, Rekognition, CloudWatch · Google Cloud Vision API, Cloud Functions, Cloud Storage, Cloud Run · Python, OpenCV, Pillow · Git, GitHub Actions

Hybrid IT Environment Setup (Azure + VirtualBox)

Deployed a main server on **Microsoft Azure** and branch servers in VirtualBox; configured secure Active Directory-based authentication (AD DS, DNS, DHCP) across hybrid environments, simulating a real-world managed-services topology.

Tools: Microsoft Azure, VirtualBox, Windows Server 2022, Active Directory

Event-Driven Data Processing Using AWS Serverless

Built a fully serverless real-time data-processing pipeline using event triggers; implemented monitoring dashboards and CloudWatch alarms for proactive alerting.

Tools: AWS Lambda, S3, EventBridge, CloudWatch · Python

Secure Enterprise VPN Network Design & Implementation

Designed and simulated a site-to-site VPN for branch offices and remote users with encrypted tunnels, strong authentication, and policy-based access control.

Tools: Cisco Packet Tracer · IPsec VPN, ACLs

Enterprise Network Design with VLANs & Remote Access

Built a multi-site enterprise network with VLANs, VTP, wireless access, and secure remote connectivity; applied QoS and inter-VLAN routing.

Tools: Cisco Packet Tracer · VLAN, VTP, OSPF, WPA2

Windows Server Administration & User Management

Managed a Windows Server 2022 environment with OUs, user accounts, Group Policy Objects (GPOs), and RBAC — directly aligning with Azure AD and ITSM operational practices.

Tools: Windows Server 2022, Active Directory, GPMC

Digital Forensics Investigation & Evidence Recovery

Recovered and analysed deleted data from USB drives; benchmarked tool performance and produced structured forensic reports.

Tools: Autopsy, ProDiscover Basic, FTK Imager

Bank Management System – Python OOP CLI Application

JOB EXPERIENCE

Student Mentor

Jun 2024 – Sep 2025

Melbourne Institute of Technology · Melbourne, VIC

- Acted as a bridge between faculty and students, providing technical guidance on networking and cybersecurity labs — developing customer-focused communication and stakeholder management skills directly applicable to a managed-services role.
- Simplified advanced security and networking frameworks for diverse cohorts, demonstrating strong written and verbal communication in technical and non-technical contexts.
- Mentored across 10 core units (Networking, Programming, Project Management), enhancing student engagement and subject mastery while managing competing priorities independently.

CERTIFICATIONS & TRAINING

- ▶ Web Designing with HTML – GitHub
- ▶ Forage Mastercard Cybersecurity Virtual Experience – Phishing Awareness Campaign Design
- ▶ CompTIA Security+ (*In Progress*)
- ▶ Microsoft Azure Fundamentals (AZ-900) (*Pursuing*)

PERSONAL ATTRIBUTES

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| ✓ Strong analytical and problem-solving skills with a security-first mindset | ✓ Customer-focused with a professional and adaptable communication style |
| ✓ Excellent teamwork and collaboration in technical and non-technical contexts | ✓ Genuine curiosity and eagerness to learn new technologies — particularly Azure |
| ✓ Self-motivated; able to work independently and manage multiple priorities | ✓ Proactive approach to identifying, mitigating, and documenting risks |

LANGUAGES

English – Professional Hindi – Professional Telugu – Native